CS23431 - OPERATING SYSTEM

EXP 4: Round Robin Algorithm Using C

QUESTION :

Write a C program to implement the Round Robin (RR) CPU Scheduling Algorithm. The program should accept the number of processes, their Arrival Times (AT), Burst Times (BT), and a Time Quantum (TQ). The CPU should rotate through the processes in the ready queue, granting each a maximum of one TQ of execution time. Calculate the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process and the overall averages.

CODE:

```c
#include <stdio.h>
int main() {
   int n, tq, i;
   printf("Enter number of processes: \n");
   scanf("%d", &n);
   printf("Enter Time Quantum: \n");
   scanf("%d", &tq);
   int AT[n], BT[n], RT[n], CT[n], TAT[n], WT[n];
   for(i = 0; i < n; i++) {
       printf("Enter AT and BT for P%d: \n", i + 1);
       scanf("%d %d", &AT[i], &BT[i]);
       RT[i] = BT[i];
   }
   int completed = 0, current_time = 0;
   float totalTAT = 0, totalWT = 0;
   while(completed < n) {
       int executed = 0;
       for(i = 0; i < n; i++) {
           if(AT[i] <= current_time && RT[i] > 0) {
               executed = 1;
               if(RT[i] > tq) {
                   current_time += tq;
                   RT[i] -= tq;
               } else {
                   current_time += RT[i];
                   RT[i] = 0;
                   CT[i] = current_time;
                   completed++;
```

```
                }
            }
        }
        if(executed == 0)
            current_time++;
    }
    for(i = 0; i < n; i++) {
        TAT[i] = CT[i] - AT[i];
}
```

Output:

Custom Test — Success

Input :

```
3
2
0 5
1 3
2 1
```

Expected Output :

```
Enter number of processes:
Enter Time Quantum:
Enter AT and BT for P1:
Enter AT and BT for P2:
Enter AT and BT for P3:

Average Turnaround Time: 6.33
Average Waiting Time: 3.33
```

Output :

```
Enter number of processes:
Enter Time Quantum:
Enter AT and BT for P1:
Enter AT and BT for P2:
Enter AT and BT for P3:

Average Turnaround Time: 6.33
Average Waiting Time: 3.33
```